BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 140 · Utility Software

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does operating-system file management do?

Q2. Why split network protocols into layers?
A. All layers must be identical
B. No interfaces are needed
C. Every packet becomes encrypted
D. Each layer has a job

Q3. “Encryption alone guarantees files cannot be stolen or deleted.” Correct this. Explain why.

Q4. A fake useful app locks files for a ransom. “Malware is any app that uses lots of RAM.” Correct this. How does this malware cause harm?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Utility software — Carries out system maintenance or security tasks alongside the operating system.

Encryption utility — Protects stored data by transforming it into ciphertext.

Compression utility — Reduces file sizes for storage or transmission.

Defragmentation — Reorganises file fragments on spinning magnetic disks; not normally beneficial for SSDs.

New learning

Utility software: Carries out system maintenance or security tasks alongside the operating system.

Encryption utility: Protects stored data by transforming it into ciphertext.

Compression utility: Reduces file sizes for storage or transmission.

Defragmentation: Reorganises file fragments on spinning magnetic disks; not normally beneficial for SSDs.

Worked example

A school has a slow magnetic HDD with heavily fragmented files. Explain defragmentation.

1. Explain file fragments may be spread across disk positions.

2. Defragmentation rearranges related blocks contiguously.

3. This can reduce mechanical seek movement.

4. Avoid recommending it as a routine performance fix for SSDs.

We Do · Practise with guidance

Guided 1

A teacher must email a folder of large documents. Select a utility.

Worked method: Note lossless methods when exact files are required.

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Write the calculation for saving storage when a file is compressed.

```
originalMB = 500
compressedMB = 350
saved = ...
print(saved)
```

write

Question 1

State a purpose of encryption utility software. [2 marks]

Question 2

Explain why defragmentation can improve HDD access. [3 marks]

Question 3

Explain why lossless compression matters for source code. [3 marks]

Exit Ticket · Answer before checking

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

Defragmenting an SSD routinely improves it in the same way as an HDD.

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. It manages files and folders: naming, saving, copying, moving and deleting them.

Q2. D — Each layer has a job

Q3. Without the key, an attacker cannot easily read ciphertext. Encryption does not stop theft or deletion.

Q4. Malware is made to harm, disrupt or gain access without permission. Different types work in different ways and can affect data access, privacy or availability.

I Do / We Do pages · Guided practice

Q1. Note lossless methods when exact files are required.

You Do pages · Code First

```
originalMB = 500
compressedMB = 350
saved = originalMB - compressedMB
print(saved)
```

Expected result: 150

You Do pages · Independent questions

Q1. Converts readable data into ciphertext to restrict unauthorised reading.

Q2. It places related fragments closer together, reducing mechanical seek time.

Q3. Original bytes must be recovered exactly; lossy removal could corrupt instructions.

Exit Ticket · Model answer

False

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question: